DoA Estimation Performance of UCAs with Reduced Number of Sensors using Phase-Mode Transformation and Small Sample Support

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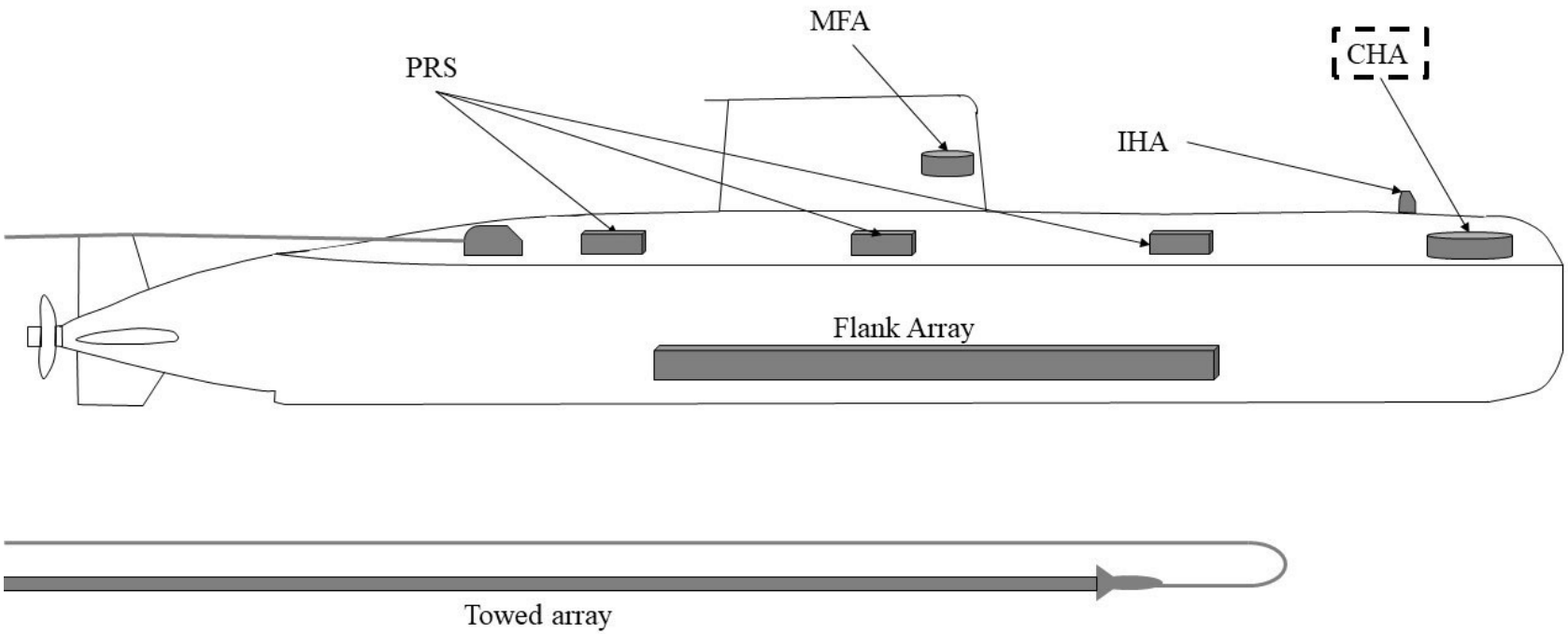
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Introduction

Aspects of Submarine Operation



Simplified set of acoustic sensor arrays (hydrophones and transducers) of a conventional submarine.

- Fast solutions in DoA estimation are desirable due to the presence of transient and fast-moving sources.
- Circular/cylindrical arrays are the centerpieces of sonar systems on any submarine.
- Greater focus on azimuthal DoA estimation, as passive ranging is heavily reliant on it.

Motivations

- Unify the works of several previous authors on UCA beamspace transform.
- Assess the effects of our 2-step uncoupled 2-D DoA approach, employing two distinct versions of MUSIC algorithm.
- Evaluate the effects of spatial aliasing and subspace leakage, in UCA 2-D DoA, both individually and simultaneously, as well as respective mitigation techniques.
- Apply these techniques to a simulated underwater environment, with real-life signal and multipath propagation, reflecting characteristics of the acoustic environment of a submarine.

Contributions

- Evaluated the performance of small sample support, beamspace, 2-D DoA estimation in an underwater simulation, using real-life signal from a cruise ship and motorboat and simulated multipath propagation.
- Obtained important results of leakage minimization and Toeplitz conditioning in the presence of correlated signals (multipath propagation).
- Successfully applied a joint phase modes incorporation-leakage minimization algorithm to assess the problem of simultaneous reduced spatial and temporal sampling.

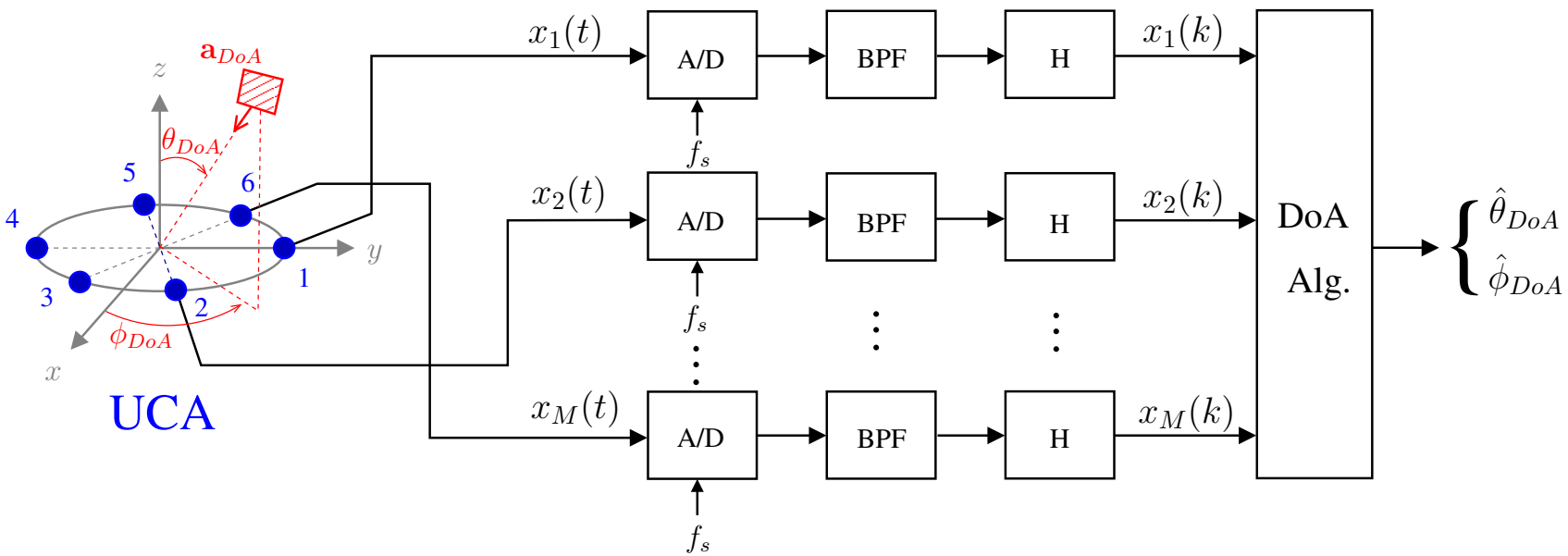
Contributions

- This work resulted in the article "DoA Estimation Performance of UCAs with Reduced Number of Sensors using Phase-Mode Transformation and Small Sample Support", published in the 12th IEEE Sensor Array and Multichannel Signal Processing (SAM) Workshop, in Trondheim, Norway.



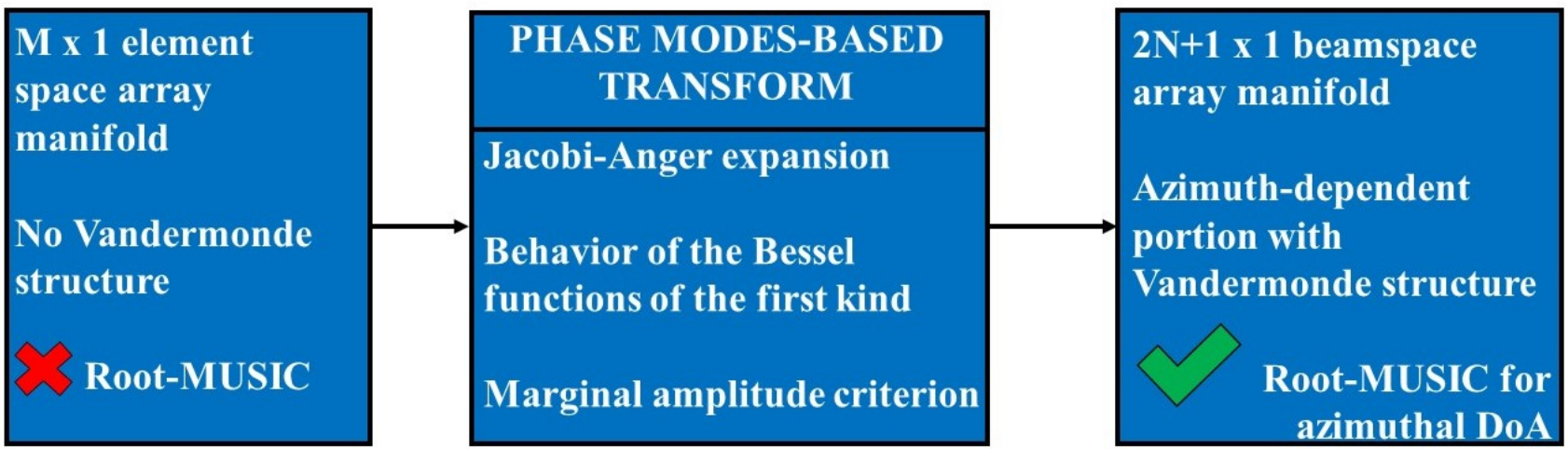
Element-space and Beamspace Uniform Circular Array Signal Model

Phase Modes-based Beamspace UCA Signal Model

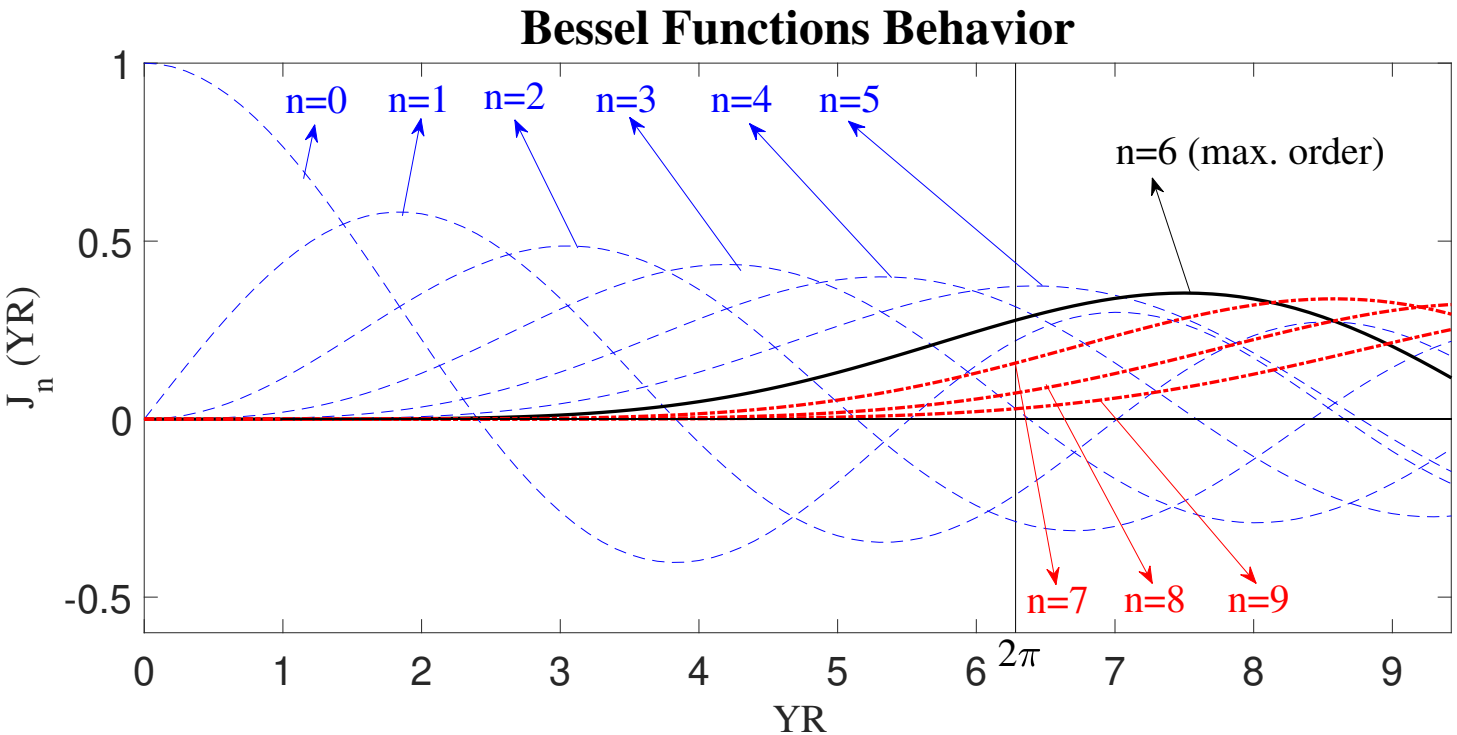


An example of an acoustic DoA estimation system with a UCA.

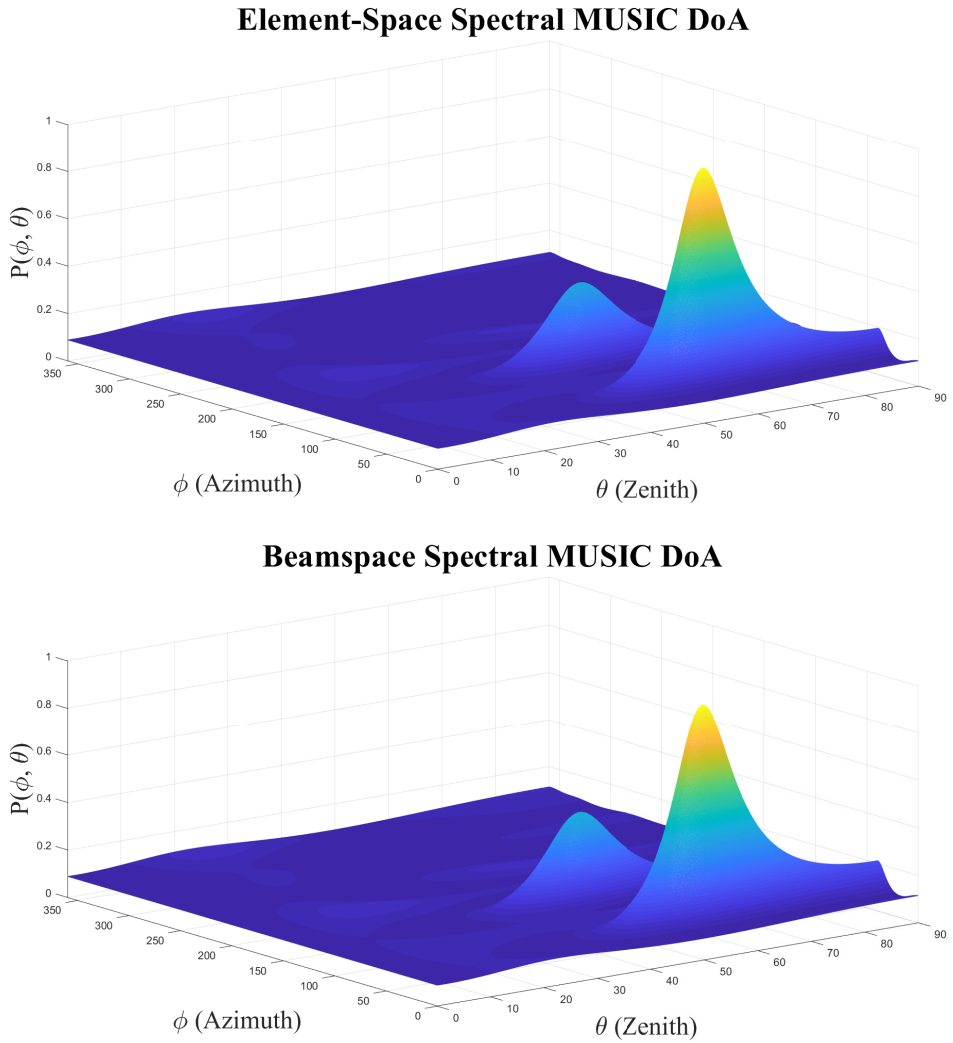
Phase Modes-based Beamspace UCA Signal Model



Phase modes-based beamspace transform diagram.

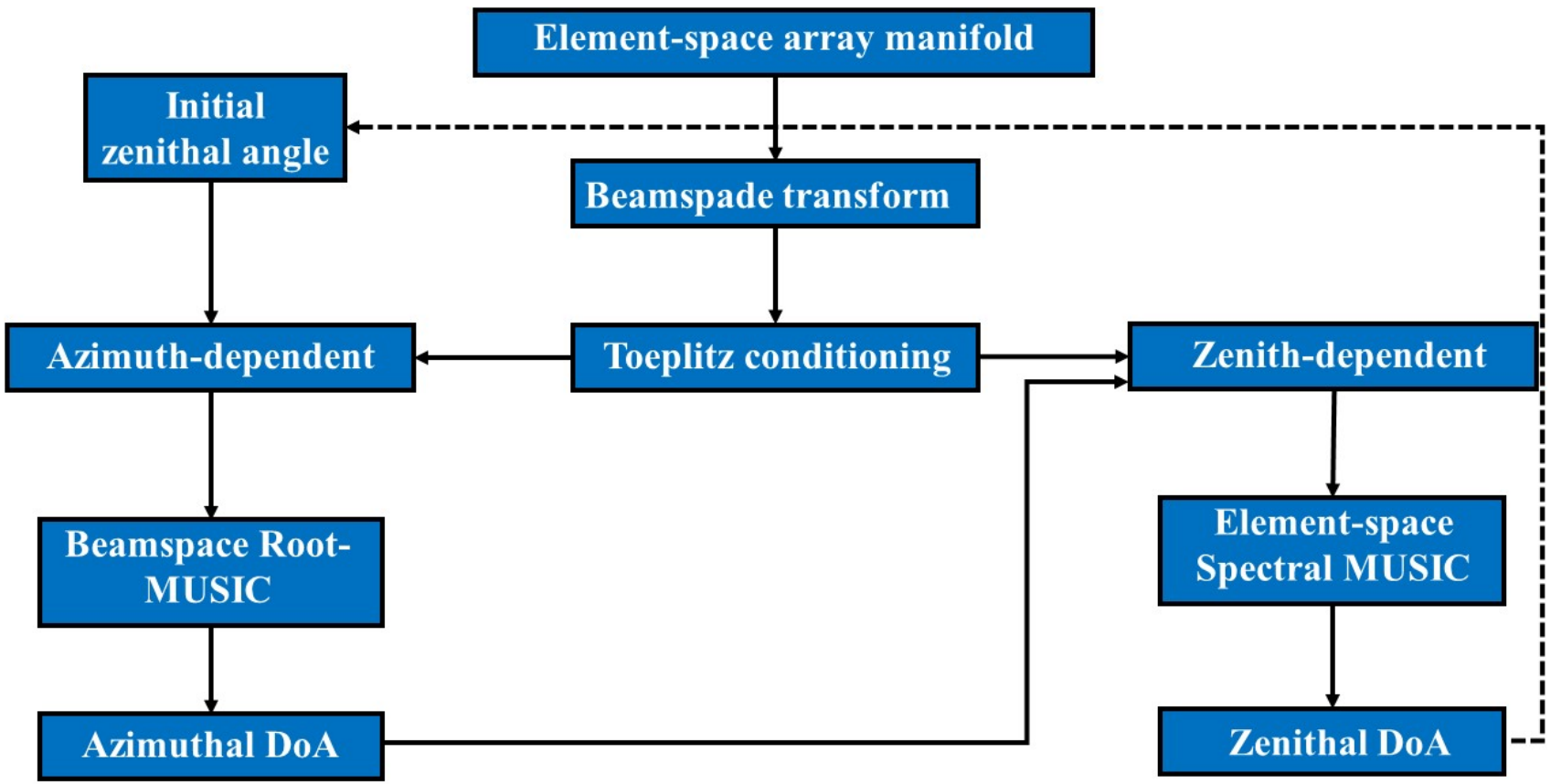

Bessel Functions Behavior
n=0
n=1
n=2
n=3
n=4
n=5
n=6 (max. order)
n=7
n=8
n=9
J_n (YR)
YR
2π
1
0.5
0
-0.5
0
1
2
3
4
5
6
7
8
9

Element-Space and Beamspace Spectral MUSIC Comparison

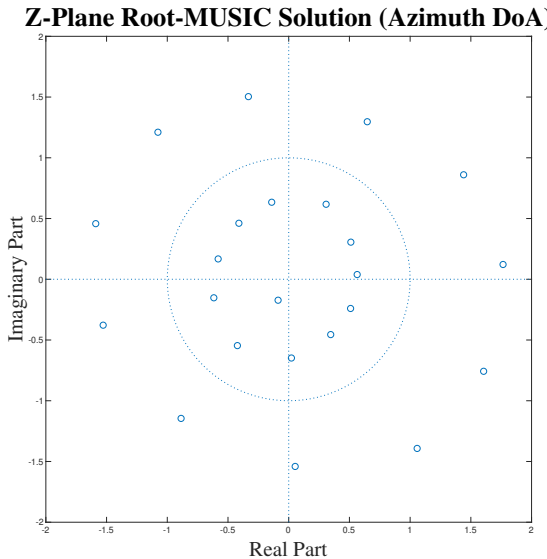


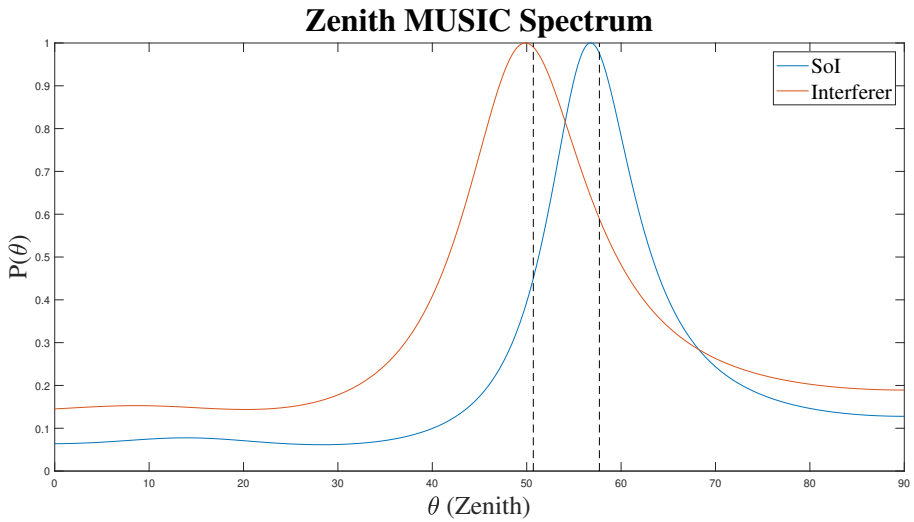
Element-space and beamspace MUSIC (10 snapshots).

Decoupled Beamspace Root-MUSIC and Spectral MUSIC DoA Estimation

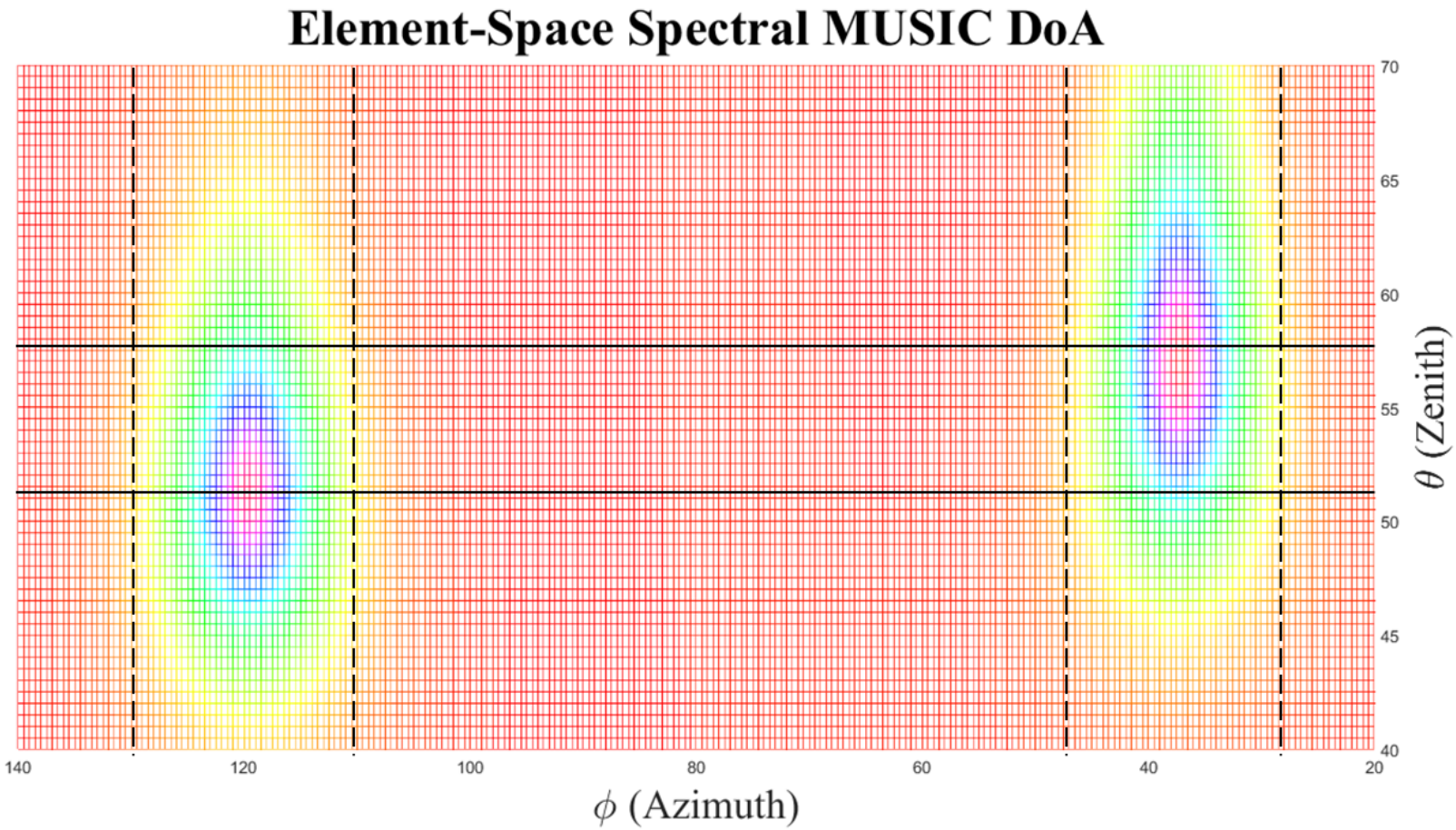


Our UCA 2-D DoA approach diagram.

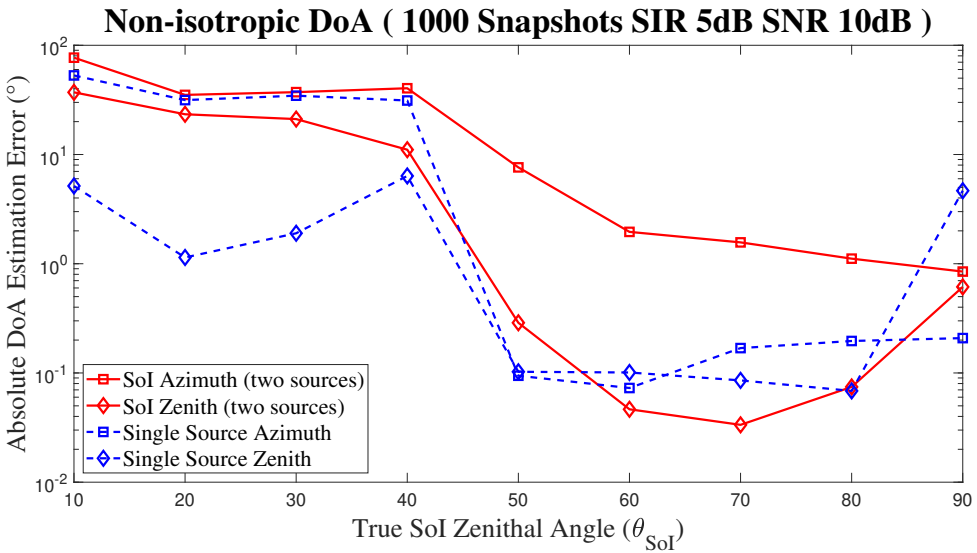


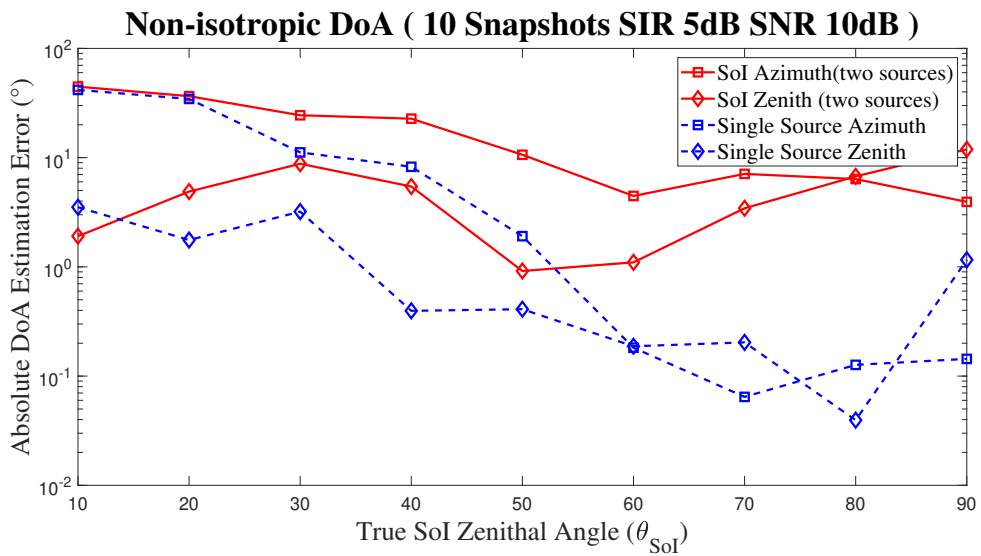


Root-MUSIC azimuth and 1-D search MUSIC zenith solutions.



Visual representation of the effects of azimuth DoA in zenith DoA estimations.

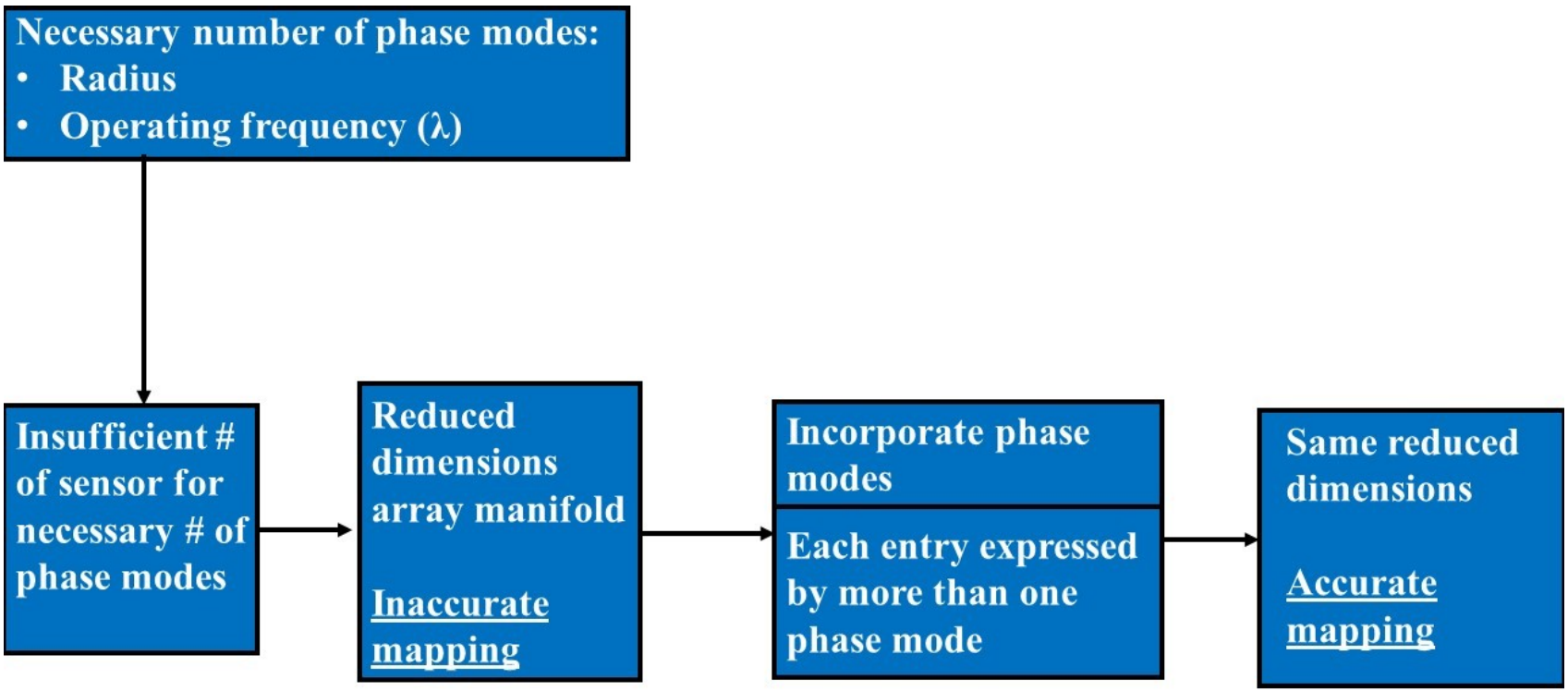




$$\theta_{SoI} \in [10°, 90°]$$

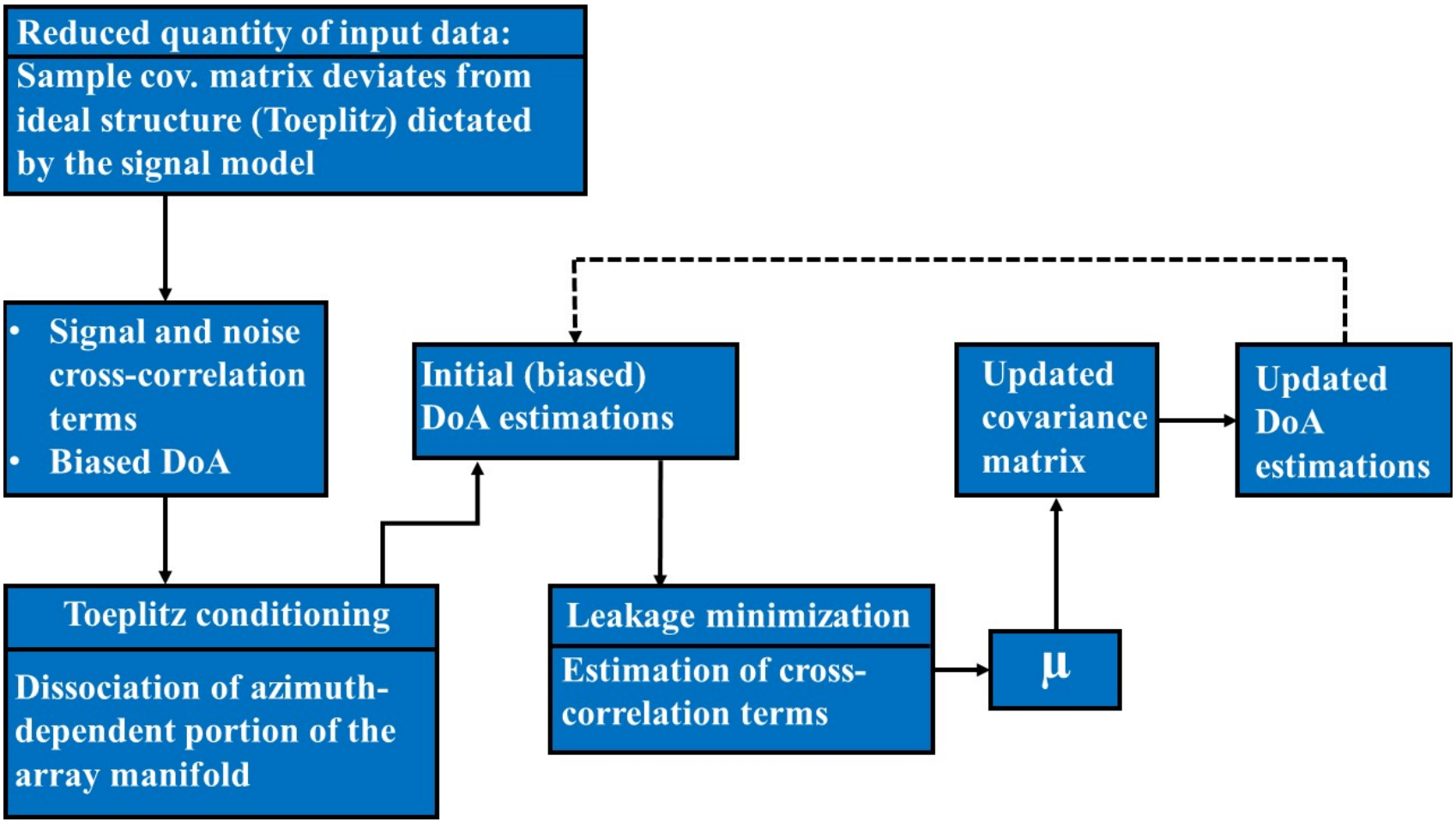
Reduced Spatial and Temporal Sampling DoA Estimation

DoA Estimation With Reduced Quantity of Sensors



Spatial aliased DoA estimation: additional phase modes incorporation diagram.

Small Sample Support: Toeplitz Conditioning and Leakage minimization

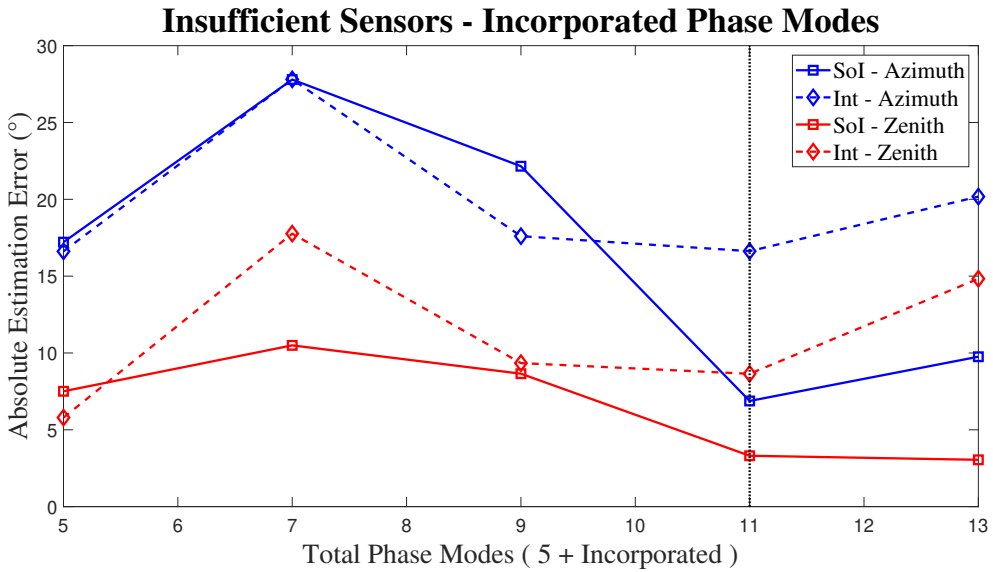


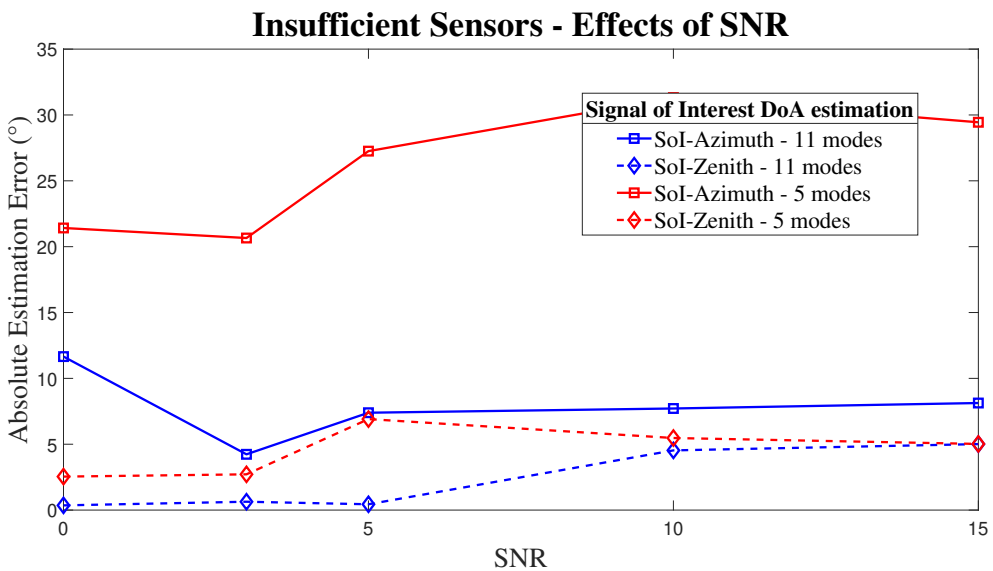
Small sample support DoA estimation diagram.

Experimental Results

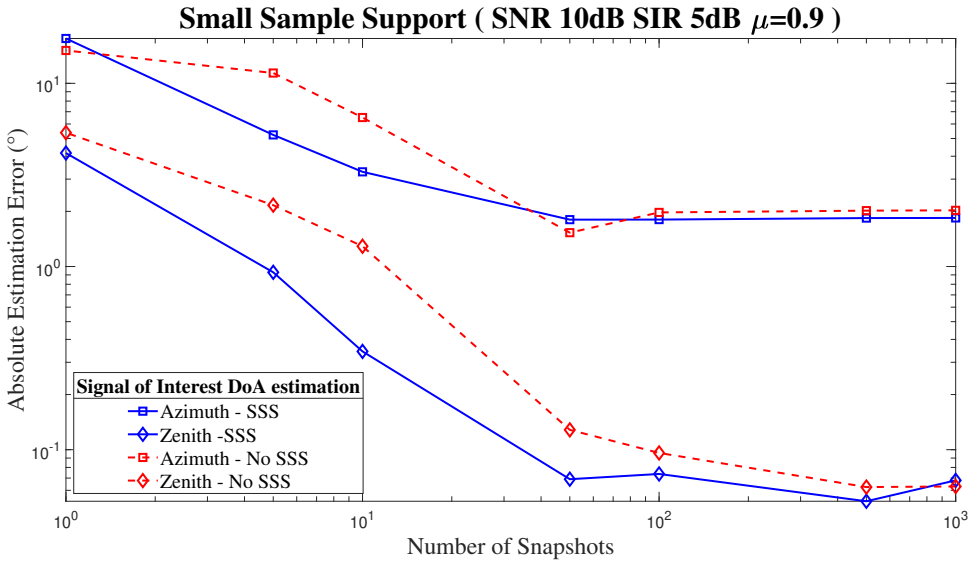
Experiments Paremeters

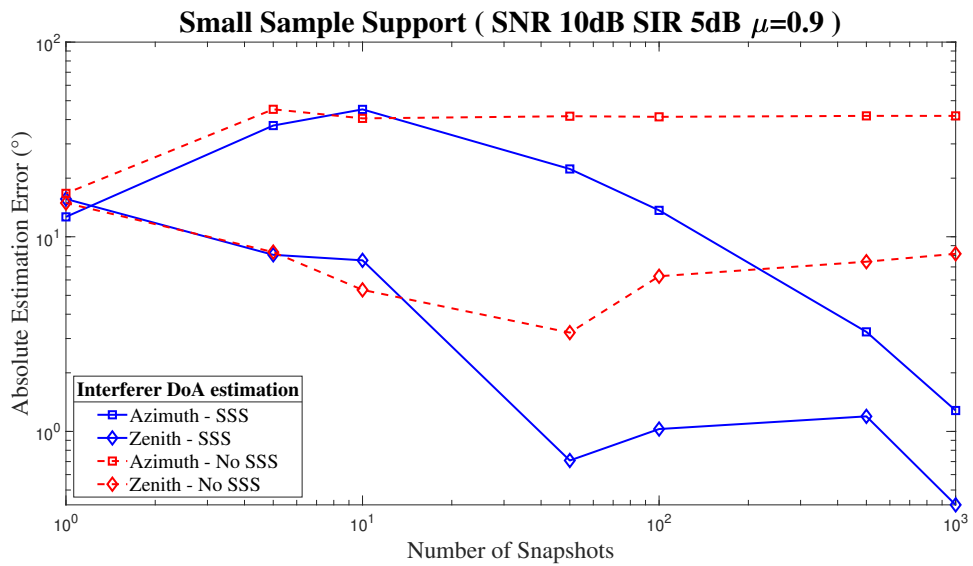
- 16 sensors, 11 phase modes, and radius $R = \lambda$. (6 sensors for sparse UCA).
- Two stochastic, uncorrelated narrowband sources with angular positions: $\phi_{Sol} = 37.9°$, $\theta_{Sol} = 57.7°$, $\phi_{Int} = 120.5°$ and $\theta_{Int} = 50.7°$.
- Sample support from 1,000 to a single snapshot.
- Underwater simulation: 32 sensors, 15 phase modes, and central frequency 1 kHz.
- Two real-life source signals with angular positions: $\phi_{CS} = 23.1°$, $\theta_{CS} = 48.5°$ and $\phi_{MB} = 142°$, $\theta_{MB} = 34.7°$. SIR 5 dB and SNR 15 dB.
- Simulated multipath propagation.



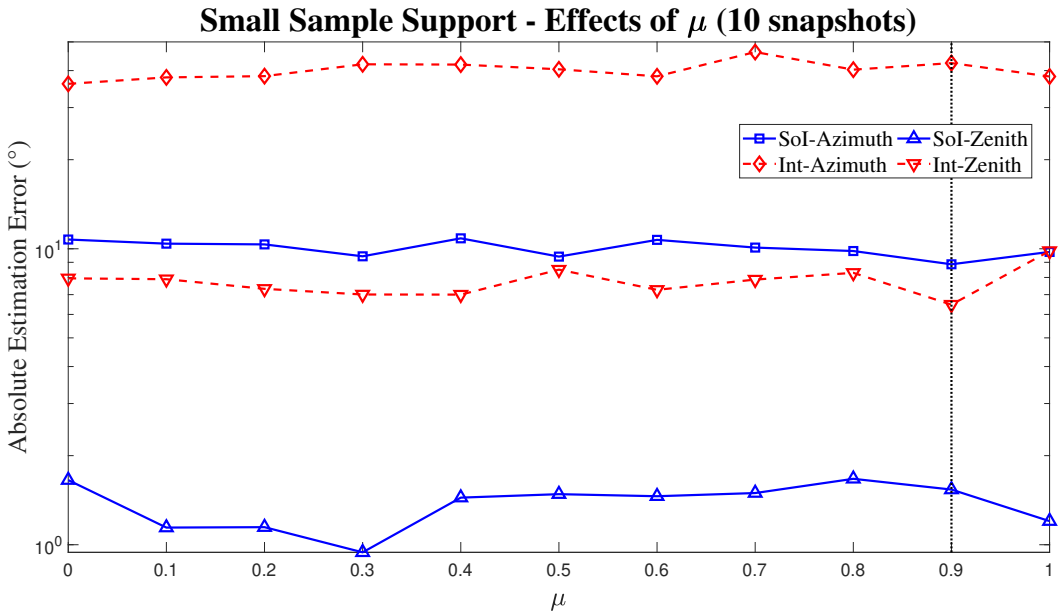


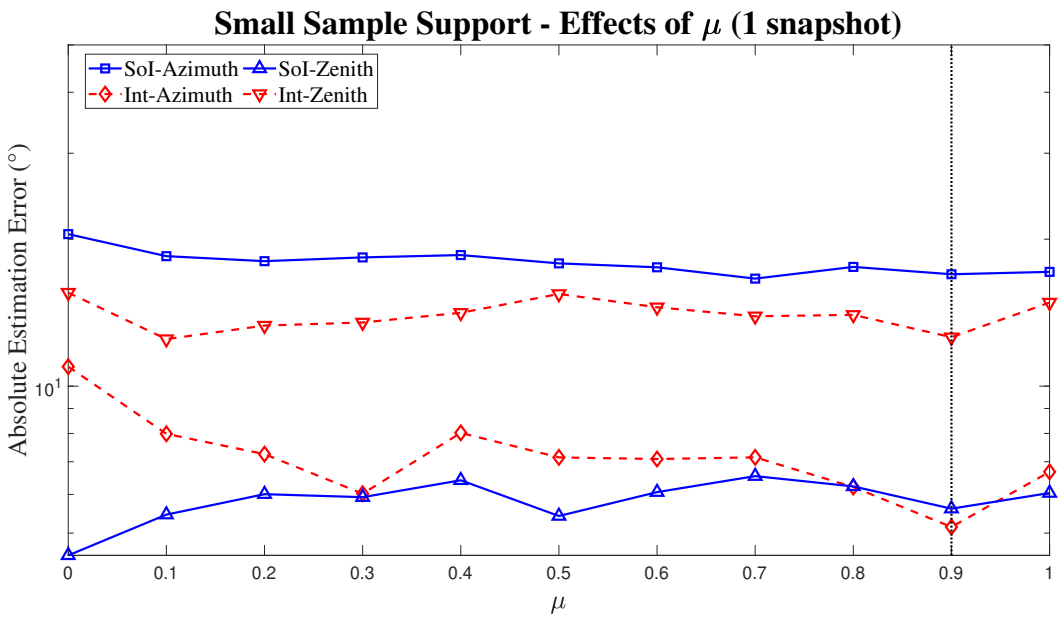
Incorporated phase modes (initial 5 up to 13).

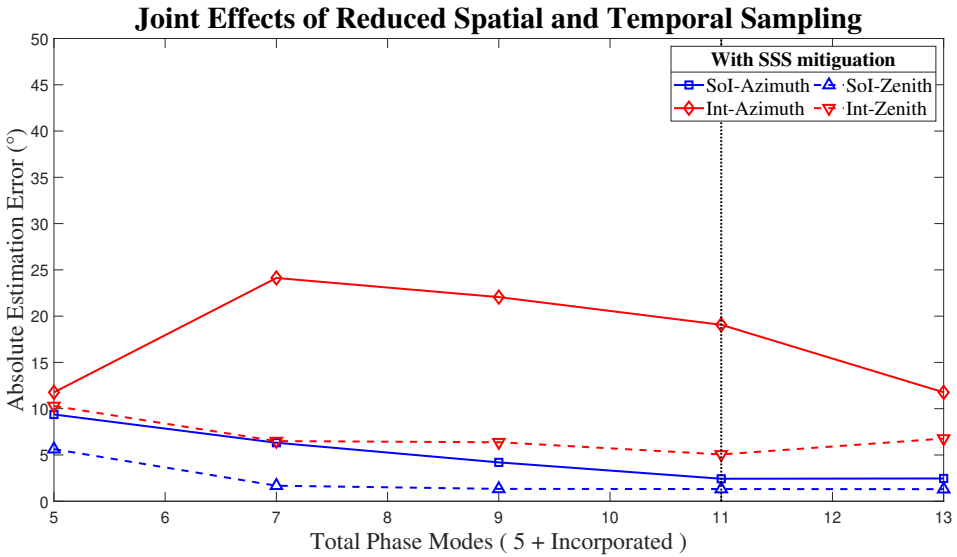


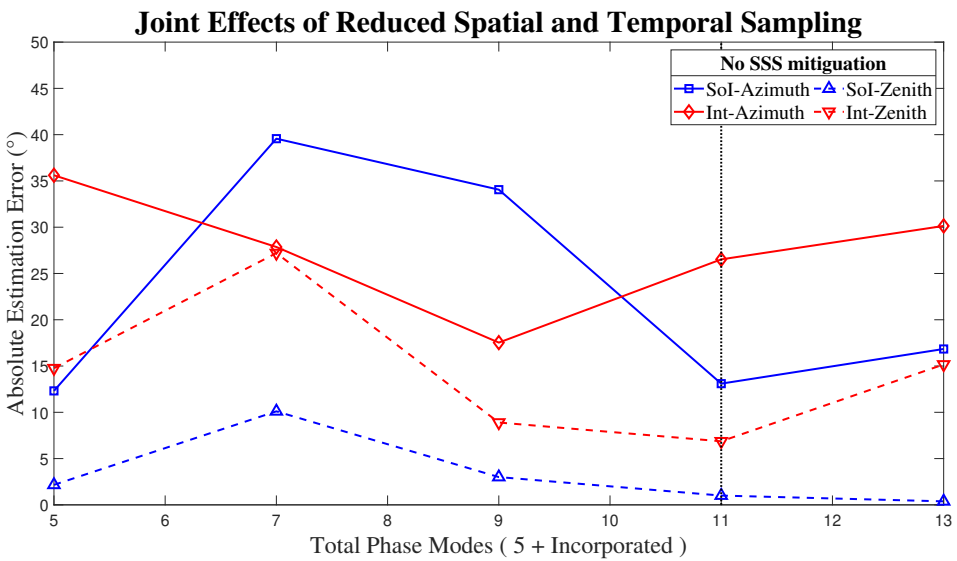


Presence (SSS) and absence (no SSS) of mitigating techniques.

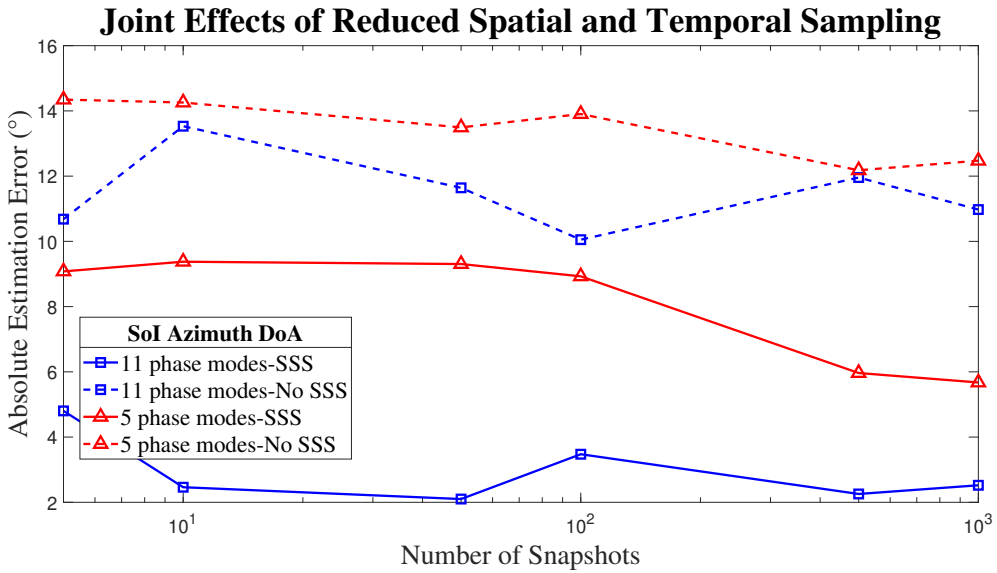

Small Sample Support - Effects of μ (10 snapshots)
SoI-Azimuth
SoI-Zenith
Int-Azimuth
Int-Zenith
Absolute Estimation Error (°)
10^1
10^0
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
μ

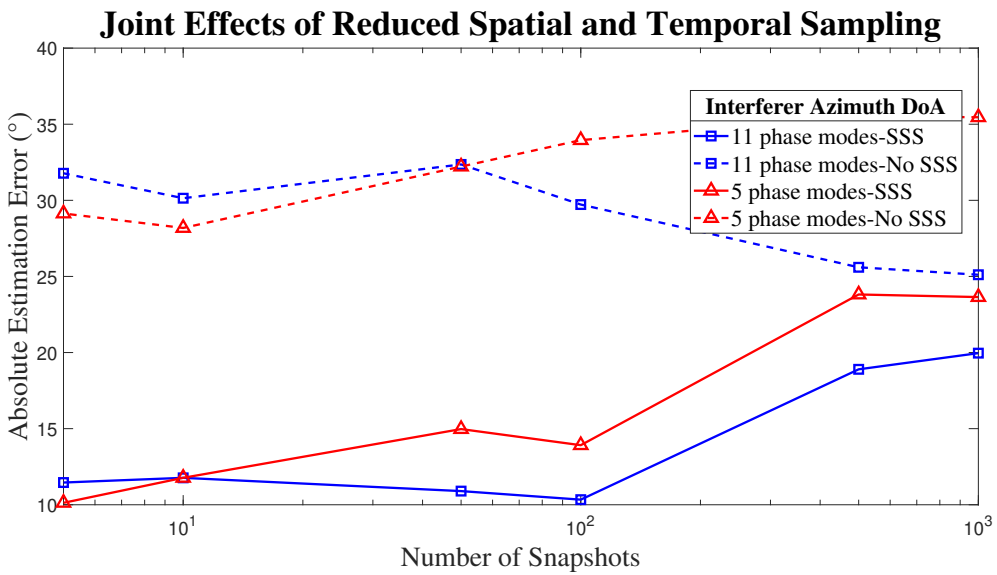

Small Sample Support - Effects of μ (1 snapshot)
SoI-Azimuth
SoI-Zenith
Int-Azimuth
Int-Zenith
Absolute Estimation Error (°)
10^1
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
μ



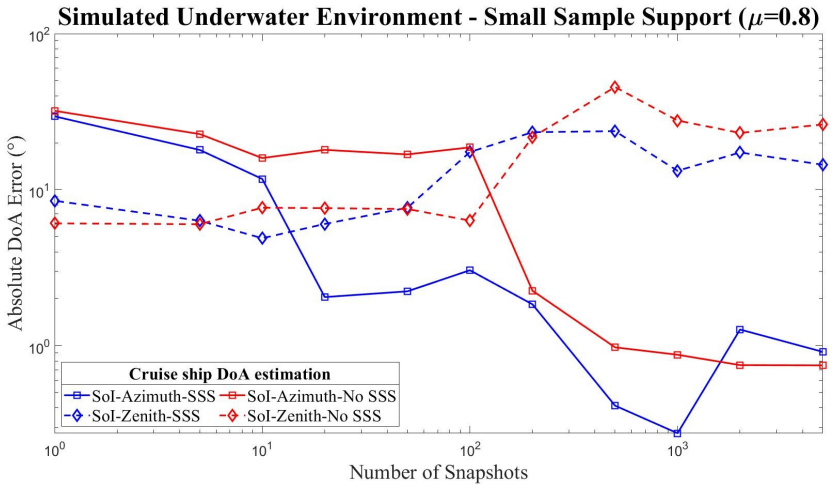


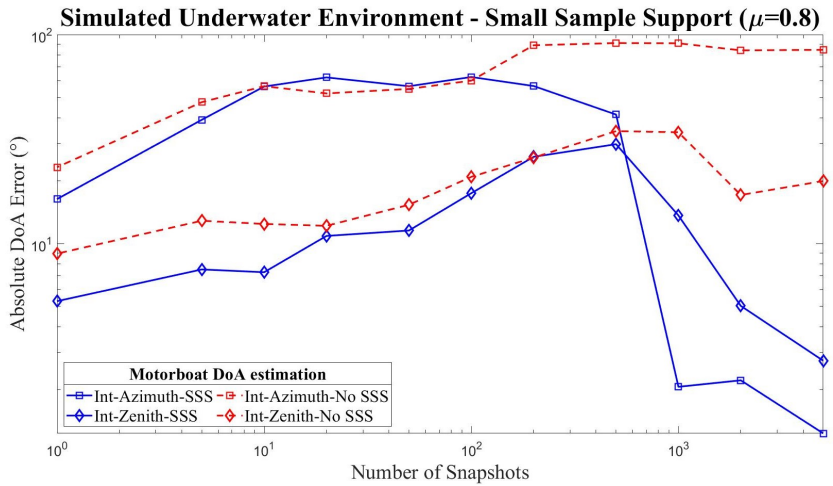
Incorporating phase modes and SSS (10 snapshots).

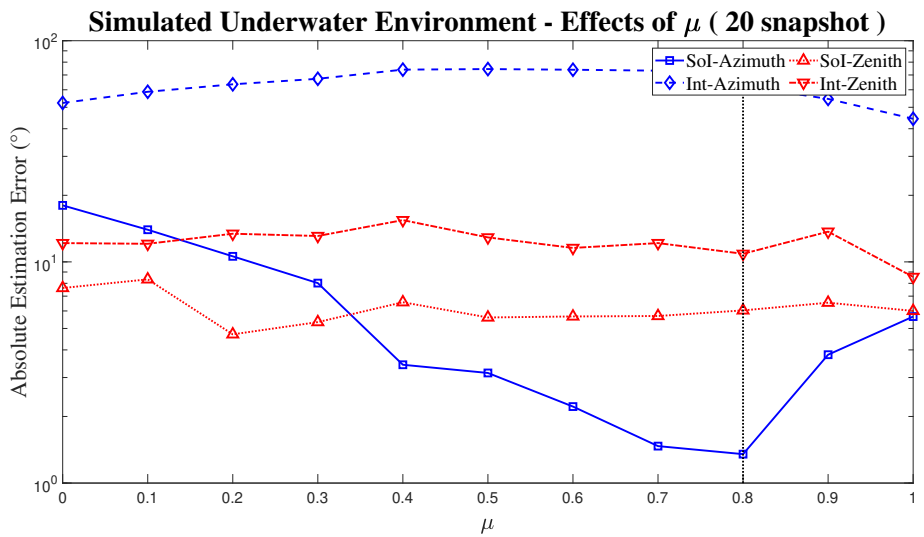




Beamspace mapping and Leakage Minimization ($\mu = 0.8$).







Decreasing sample support in UW simulation ($\mu = 0.8$).

Conclusion

Final Remarks

- Methods presented are directed to beamspace Root-MUSIC approach and, consequently, presents greater improvements in azimuthal DoA estimation.

- Incorporating additional phase modes, up to an optimum value, improves considerably the DoA estimation. In our simulations, with an UCA of six sensors, we found that the optimum beamspace mapping was through 11 phase modes.

- Matrix conditioning could mitigate biased covariance matrices, with $\mu = 0.9$ as the optimum value for our simulations. This value corroborates our good estimation of cross-correlation.

Final Remarks

- Regarding simultaneous reduced temporal and spatial sampling, Toeplitz Conditioning cannot be used with sparse UCA.

- The joint behavior of incorporating phase modes and Leakage Minimization ($\mu = 0.8$) follows partially the isolated ones. The later presents more pronounced overall effects.

- In the underwater environment simulation, the effects of multipath propagation, akin to correlated signals, are present. Although not addressed directly, Toeplitz Conditioning and Leakage Minimization ($\mu = 0.8$) end up yielding favorable results in this additional limiting condition.

Future Works

- Apply Phase Modes-based beamspace transform to suitable volumetric geometries: cylindrical and spherical arrays.
- Extend UCA beamspace 2-D DoA to passive ranging estimation via Horizontal Direct Passive Ranging (HDPR).
- Employ beamspace DoA estimation with volumetric arrays in passive ranging with multipath propagation via Vertical Direct Passive Ranging (VDPR).
- Extend beamspace 2-D DoA with reduced spatial and temporal sampling to volumetric geometries and passive ranging applications.

Thank you for your time